

Two-Photon Physics at the $LH\mu C$

High-energy $\gamma\gamma$ interactions in μp collisions

Hamzeh Khanpour
AGH University of Krakow

(on behalf of HELP4CERN Study Group)

Ultra-cold muons for the LHC, AGH Krakow, 24-26 September 2026

- 1 LH μ C and the $\gamma\gamma$ opportunity
- 2 Equivalent Photon Approximation and $\gamma\gamma$ luminosity
- 3 Standard-Model $\gamma\gamma$ processes
- 4 Precision / EFT probes: τ moments and gauge couplings
- 5 Heavy final states
- 6 Experimental opportunities and outlook

Why a muon-hadron collider for two-photon physics?



- A muon-hadron collider combines a clean lepton-hadron initial state with a multi-TeV centre-of-mass energy.
- Replacing the electron beam by a muon beam strongly relaxes the synchrotron-radiation limitation, making much higher lepton-beam energies possible.
- The μp environment remains experimentally cleaner than pp for many exclusive and semi-exclusive final states: negligible pile-up is a central advantage of the lepton-hadron concept.
- Both beams radiate photons. Their electromagnetic fields therefore create a high-energy $\gamma\gamma$ collider *inside* the μp machine.

A high-energy $\gamma\gamma$ collider!

LH μ C combines a high-energy photon source from the muon beam with the electromagnetic photon field of the proton, opening a distinctive high-energy $\gamma\gamma$ programme.

LH μ C@CERN: benchmark working points



- The present talk focuses on the two-photon physics case of the LH μ C, using the current working benchmarks:
- The first option: $E_\mu = 500$ GeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 3.7$ TeV, with a $\mathcal{L}_{\text{int}} = 250$ fb $^{-1}$.
- **The nominal baseline:** $E_\mu = 1$ TeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 5.3$ TeV, with a $\mathcal{L}_{\text{int}} = 1000$ fb $^{-1}$.
- The second option: $E_\mu = 2$ TeV, $E_p = 7$ TeV, $\sqrt{s} \simeq 7.5$ TeV, with a $\mathcal{L}_{\text{int}} = 1000$ fb $^{-1}$.

LH μ C physics message

These working points are sufficient to demonstrate the main message of LH μ C: already at the 500 GeV, LH μ C enters the multi-TeV regime and becomes highly competitive for $\gamma\gamma$ physics, Higgs, top, electroweak, and representative BSM studies.

Benchmark WPs

Machine	Benchmark
LHeC	$E_e = 20, 50$ GeV
LH μ C (option 1)	$E_\mu = 500$ GeV
LH μ C (baseline)	$E_\mu = 1000$ GeV
LH μ C (option 2)	$E_\mu = 2000$ GeV
Configuration	Value
$\sqrt{s}_{\mu p}$ (option 1)	$\simeq 3.7$ TeV
$\sqrt{s}_{\mu p}$ (baseline)	$\simeq 5.3$ TeV
$\sqrt{s}_{\mu p}$ (option 2)	$\simeq 7.5$ TeV
$\mathcal{L}_{\text{int}}$ (option 1)	250 fb $^{-1}$
$\mathcal{L}_{\text{int}}$ (baseline)	1000 fb $^{-1}$
$\mathcal{L}_{\text{int}}$ (option 2)	1000 fb $^{-1}$

Representative benchmark cross sections



Why is this relevant for $\gamma\gamma$ physics?

Moving from the LHeC to the LH μ C strongly increases the available lepton-hadron centre-of-mass energy and opens the multi-TeV regime. The same energy lever arm is crucial for two-photon physics: the photon spectrum from the high-energy muon beam becomes significantly harder, extending the accessible $W_{\gamma\gamma}$ range. This enhances the reach for heavy $\gamma\gamma$ final states and increases sensitivity to new interactions in the high-mass tails.

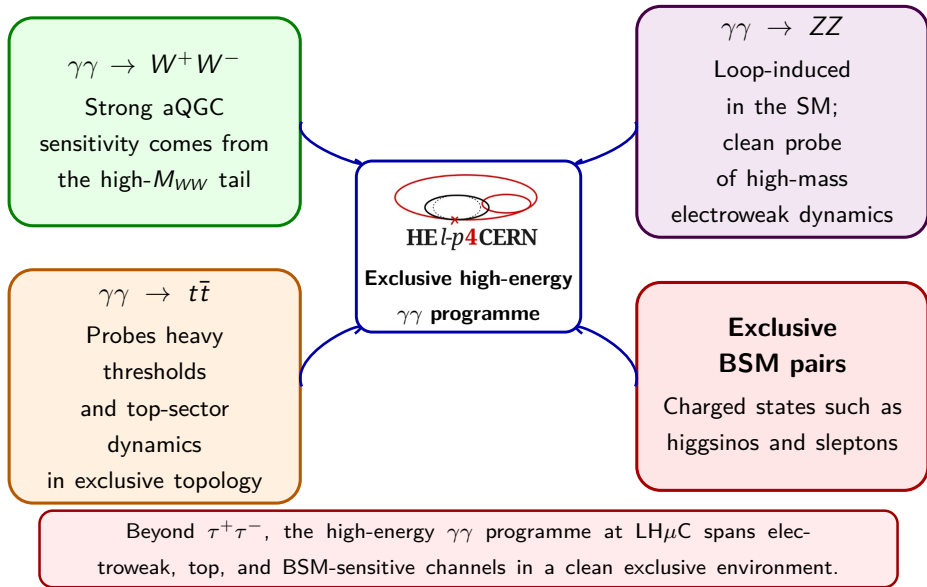
Benchmark inclusive cross sections

Process	20 GeV	50 GeV	500 GeV	1 TeV
	LHeC	LHeC	LH μ C	LH μ C
$\sigma(\ell p \rightarrow \nu_\ell H + \text{jet} + X)$ [fb]	52	164	438	731
$\sigma(\ell p \rightarrow \ell H + \text{jet} + X)$ [fb]	5.6	19	135	221
$\sigma(\ell p \rightarrow \ell t\bar{t} + X)$ [pb]	0.0015	0.0323	1.03	2.23
$\sigma(\ell p \rightarrow \nu_\ell t + X)$ [pb]	0.55	2.89	21.26	36.77

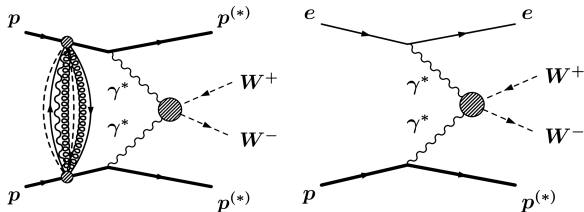
Key comparisons

Relative to the 50 GeV LHeC benchmark, the nominal 1 TeV LH μ C baseline enhances the benchmark inclusive rates by approximately: ~ 4.5 for charged-current Higgs production, ~ 12 for neutral-current Higgs production, ~ 13 for single-top production, and ~ 69 for $t\bar{t}$ production.

Exclusive $\gamma\gamma$ channels: electroweak, top, and BSM reach



Two-photon production in lepton-proton (ℓp) scattering



pp: two hadronic beams

ℓp : one hadronic beam

Two-photon searches at the LHC already demonstrate the physics potential of exclusive and semi-exclusive $\gamma\gamma$ interactions. Moving to a lepton-hadron collider removes one of the two hadronic initial states and therefore considerably simplifies the event environment.

Factorized $\gamma\gamma$ picture

$$\ell \rightarrow \ell + \gamma^*, \quad p \rightarrow p^{(*)} + \gamma^*,$$

$$\gamma\gamma \rightarrow X,$$

$$\boxed{\ell p \rightarrow \ell + X + p^{(*)}}$$

The incoming lepton and proton act as sources of quasi-real photons. The hard central system X is then produced through $\gamma\gamma$ fusion.

From LHeC to LH μ C

At the LH μ C, the higher lepton-beam energy produces a harder equivalent-photon spectrum, extending the accessible $W_{\gamma\gamma}$ range and strengthening the reach for heavy final states and high-mass EFT/BSM effects.

$\gamma\gamma$ at ℓp : production topologies and experimental handles



Photon-emission topologies

• Elastic

$$p \rightarrow p + \gamma^*$$

The proton remains intact. Its electromagnetic form factors $F_E(Q^2)$ and $F_M(Q^2)$ determine the elastic photon flux.

• Proton-dissociative

$$p \rightarrow X_p + \gamma^*, \quad M_N > m_p.$$

The proton breaks up into a hadronic system X_p . The photon flux is then described using proton structure functions. Allowing larger proton-dissociation masses M_N and photon virtualities Q_p^2 increases the accessible $\gamma\gamma$ luminosity.

Experimental handles

- **Low pile-up** preserves clean final states and makes rapidity-gap / exclusivity requirements particularly effective.
- **Reduced hadronic re-scattering** compared with pp : there is no second hadronic beam and hence no analogous soft-survival complication on the lepton side.
- The **scattered lepton** constrains the photon emission from the lepton side; radiative corrections are predominantly QED, including ISR.
- A **forward proton tag**, when available, can over-constrain the event kinematics and is especially valuable for final states containing missing energy.

Equivalent Photon Approximation: factorising the ℓp cross section

EPA master relation

$$\sigma_{\mu p} = \int dy_\mu dy_p \Phi_\mu(y_\mu) \Phi_p(y_p) \sigma_{\gamma\gamma}(W) = \int dW S_{\gamma\gamma}(W) \sigma_{\gamma\gamma}(W)$$

- Φ_μ and Φ_p are the equivalent-photon fluxes radiated by the two beams.
- The hard subprocess $\gamma\gamma \rightarrow X$ factorises from photon emission, in close analogy with partonic convolutions.
- $S_{\gamma\gamma}(W)$ packages the two fluxes into the effective two-photon luminosity spectrum.

Kinematic variables

$$y_\mu = \frac{E_\gamma^{(\mu)}}{E_\mu}, \quad y_p = \frac{E_\gamma^{(p)}}{E_p},$$

$$W \equiv W_{\gamma\gamma}.$$

Equivalent photon fluxes from the leptons (e or μ^+) and proton



AGH

www.agh.edu.pl

Generic EPA flux

$$\Phi_\gamma(y) = \frac{\alpha}{\pi y} \int \frac{dQ^2}{Q^2} \left[(1-y) \left(1 - \frac{Q_{\min}^2}{Q^2} \right) F_E(Q^2) + \frac{y^2}{2} F_M(Q^2) \right]$$

$$Q_{\min}^2 = \frac{M^2 y^2}{1-y}.$$

- Point-like lepton: $F_E = F_M = 1$ and $M = m_\mu$.
- The muon mass raises $Q_{\min}^2$ relative to an electron at the same photon energy fraction y .

Elastic proton emission

$$F_M = G_M^2,$$

$$F_E = \frac{4m_p^2 G_E^2 + Q^2 G_M^2}{4m_p^2 + Q^2}.$$

The dipole form factors suppress large Q_p^2 .

Proton-dissociative emission

$$F_E(Q^2) = \int \frac{dx}{x} F_2(x, Q^2), \quad F_M(Q^2) = \int \frac{dx}{x^3} F_2(x, Q^2).$$

For $p \rightarrow X_p + \gamma^*$, proton structure functions replace the elastic form factors; the flux depends on the photon virtuality Q_p^2 and the dissociative mass M_N : $M_N^2 = m_p^2 + Q^2 \left(\frac{1}{x} - 1 \right)$.

Beyond the simplest EPA: photon virtuality and exact $W_{\gamma\gamma}$ kinematics



High-energy approximation

$$W^2 \simeq y_\ell y_p s.$$

Exact elastic and inelastic kinematics

Elastic:

$$W^2 = -Q_\ell^2 + y_\ell y_p s$$

in which depends on the virtuality Q_ℓ^2 of the photon emitted by the lepton.

Proton-dissociative:

$$W^2 = -Q_\ell^2 - Q_p^2 + y_\ell y_p s + \frac{Q_\ell^2}{s} (Q_p^2 + M_N^2 - m_p^2)$$

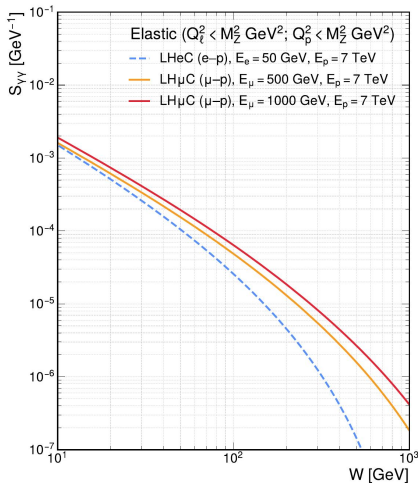
in which depend on the proton-side photon virtuality Q_p^2 and on the invariant mass M_N of the dissociative hadronic system.

- The approximation is excellent when photon virtualities are negligible relative to W^2 .
- For proton dissociation, the exact expression also depends on Q_p^2 and M_N .
- Elastic EPA agrees at percent level with matrix-element generators once the proper W definition is used; inelastic modelling is more sensitive to the dissociative phase space.

Precision lesson

At sizeable Q^2 , correct $W_{\gamma\gamma}$ reconstruction is part of the physics model, not a cosmetic refinement.

What does the $\gamma\gamma$ luminosity spectrum $S_{\gamma\gamma}$ encode?



Elastic photon-photon luminosity spectrum $S_{\gamma\gamma}$ as a function of the $\gamma\gamma$ invariant mass W , comparing the LHeC and LH μ C benchmark configurations.

$S_{\gamma\gamma}$ tells us

$S_{\gamma\gamma}(W)$ is the effective photon-photon luminosity density obtained from the convolution of the equivalent-photon fluxes of the lepton and proton. It is the universal machine-level input that determines how much luminosity is available at a given $\gamma\gamma$ invariant mass W .

- Increasing the lepton-beam energy shifts the usable $\gamma\gamma$ luminosity to larger W : the LH μ C spectra fall more slowly than the LHeC spectrum in the high-mass region.
- Already for $E_\mu = 500$ GeV, the accessible high- W region is substantially enlarged; the 1 TeV nominal option extends it further.

From luminosity to a physical process

$$\sigma_{\ell p \rightarrow \ell X p^{(*)}} = \int dW S_{\gamma\gamma}(W) \sigma_{\gamma\gamma \rightarrow X}(W).$$

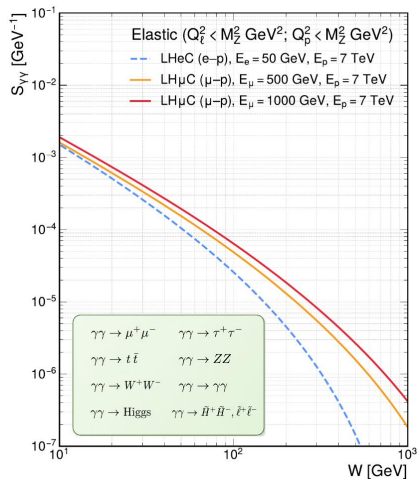
$S_{\gamma\gamma}$ is process independent: the final-state physics enters through the subprocess cross section $\sigma_{\gamma\gamma \rightarrow X}$.

The high-energy $\gamma\gamma$ physics landscape at LH μ C



Interesting HET directions

- Once the photon luminosity is folded with the relevant $\gamma\gamma$ subprocess, the LH μ C provides access to a broad and complementary two-photon programme.
- Precision QED and lepton physics: $\gamma\gamma \rightarrow \mu^+\mu^-, \tau^+\tau^-$, including anomalous τ electromagnetic moments,
- Electroweak and EFT probes: $\gamma\gamma \rightarrow W^+W^-$ and sensitivity to anomalous quartic gauge couplings,
- Rare neutral-gauge channel: loop-induced $\gamma\gamma \rightarrow ZZ$, with sensitivity to neutral quartic gauge interactions,
- Heavy charged final states: $\gamma\gamma \rightarrow t\bar{t}$ and other high-threshold processes,
- BSM opportunities: light-by-light scattering and pair production of new electrically charged particles.



Elastic $S_{\gamma\gamma}(W)$ for the LHeC and LH μ C benchmark configurations.

Benchmark channels in the LH μ C $\gamma\gamma$ programme



Collider rate

$$\sigma(\mu p \rightarrow \mu X p^{(*)}) = S_{\gamma\gamma} \otimes \sigma(\gamma\gamma \rightarrow X).$$

$\ell^+\ell^-$
calibration

$\tau^+\tau^-$
dipole
moments

W^+W^-
gauge
structure

ZZ
loop / nQGC

$t\bar{t}$
heavy
threshold

charged BSM
pairs

$\gamma\gamma$
light-by-light

H
loop-induced

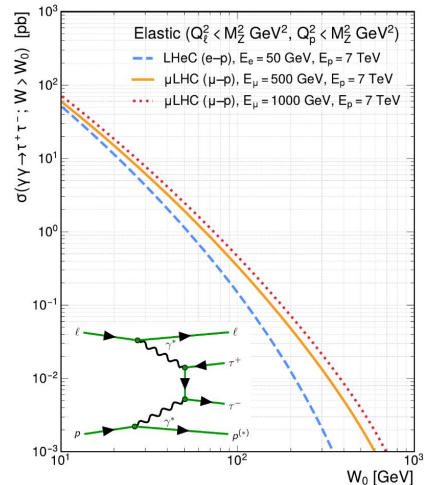
Benchmark studies highlighted in this talk

Two benchmark studies will be discussed in detail: $\gamma\gamma \rightarrow \tau^+\tau^-$, probing anomalous τ electromagnetic moments, and $\gamma\gamma \rightarrow W^+W^-$, probing anomalous quartic gauge couplings. Together they illustrate the complementary precision and EFT sensitivity of the LH μ C $\gamma\gamma$ programme.

Benchmark $\gamma\gamma$ channel: $\tau^+\tau^-$ and anomalous τ moments

LH μ C $\tau^+\tau^-$ physics message

- The elastic $\gamma\gamma \rightarrow \tau^+\tau^-$ rate at LH μ C for $W_0 = 10$ GeV is already very large, with $\sigma_{\text{el}} \simeq 58.5$ pb at 3.7 TeV, and about 70.3 pb at the nominal 5.3 TeV baseline.
- This provides a high-statistics probe of the $\gamma\tau\tau$ vertex and anomalous magnetic (a_τ) and electric (d_τ) dipole moments.
- Anomalous contributions harden the high- $M_{\tau\tau}$ tail, making the invariant-mass spectrum the key discriminating observable.
- This is one of the cleanest examples of how LH μ C combines high energy with a low-background exclusive topology.



Elastic $\gamma\gamma \rightarrow \tau^+\tau^-$ cross section above the threshold W_0 , comparing the LHeC and LH μ C benchmark configurations.

Anomalous τ electromagnetic moments: the $\gamma\tau\tau$ vertex

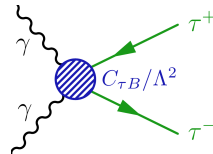


General electromagnetic vertex

$$\Gamma^\mu(q^2) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m_\tau} F_2(q^2) + \sigma^{\mu\nu}q_\nu \gamma_5 F_3(q^2),$$

$$F_2(0) = a_\tau, \quad F_3(0) = -\frac{2m_\tau}{e} d_\tau.$$

- F_1 : electric charge form factor.
- a_τ : CP-even anomalous magnetic dipole moment.
- d_τ : CP-odd electric dipole moment.
- In the SM, both are very small; any sizable deviation would signal new physics.
- $\gamma\gamma \rightarrow \tau^+\tau^-$ probes the $\gamma\tau\tau$ vertex directly.



SMEFT interpretation

Gauge-invariant dipole operators can be written as

$$\bar{L}_\tau \sigma^{\mu\nu} \tau_R H \left(\frac{C_{\tau B}}{\Lambda^2} B_{\mu\nu} + \frac{C_{\tau W}}{\Lambda^2} W_{\mu\nu} \right) + \text{h.c.}$$

After electroweak symmetry breaking, suitable real and imaginary combinations of $C_{\tau B}$ and $C_{\tau W}$ generate respectively magnetic and electric dipole deformations.

How anomalous τ moments appear in $\gamma\gamma \rightarrow \tau^+\tau^-$

Physical consequences in $\gamma\gamma \rightarrow \tau^+\tau^-$

- The anomalous magnetic moment a_τ modifies the CP-even production rate and differential distributions.
- The electric dipole moment d_τ is CP-odd; in unpolarised inclusive rates it typically enters quadratically, while dedicated CP-odd observables can provide linear sensitivity.
- Both effects become more visible in the hard kinematic regime, especially at large $M_{\tau\tau}$, where the BSM contribution grows relative to the SM baseline.
- The τ -pair channel is therefore a natural benchmark to connect high-energy $\gamma\gamma$ collisions with dipole-moment searches.

Experimental strategy before limit setting

- Use the clean $\gamma\gamma \rightarrow \tau^+\tau^-$ sample as the benchmark channel.
- Focus on observables such as

$$M_{\tau\tau}, \quad Y_{\tau\tau}, \quad p_T^\tau,$$

with the **high- $M_{\tau\tau}$ tail** providing the leading lever arm.

- Compare the measured differential spectra with the SM expectation and with benchmark deformations in a_τ and/or d_τ .
- The next step is therefore to turn these distortions into **projected constraints / limits**.

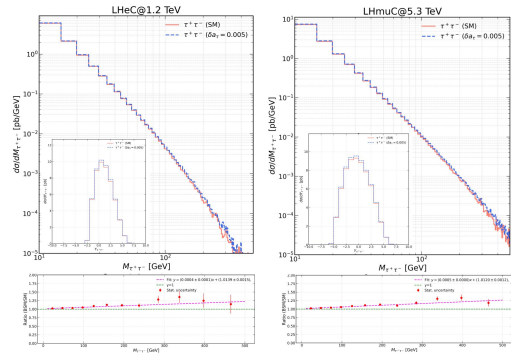
Why LHC μ C is promising

Relative to the LHeC benchmark, the LHC μ C provides a harder $\gamma\gamma$ luminosity spectrum and better statistical reach in the high- $M_{\tau\tau}$ tail, where sensitivity to anomalous dipole couplings is strongest.

Current limits and high- $M_{\tau\tau}$ sensitivity to δa_τ



- For the benchmark deformation $\delta a_\tau = 0.005$, the deviation from the SM remains small at low $M_{\tau\tau}$, but grows in the high- $M_{\tau\tau}$ tail; this is the main source of sensitivity.
- Compared with the LHeC benchmark, the nominal LH μ C setup probes a harder kinematic regime and retains better statistical reach in the tail, giving stronger sensitivity leverage on anomalous τ dipole couplings.
- These distributions therefore indicate good expected sensitivity at LH μ C, but a quantitative projected limit still requires a dedicated detector-level statistical analysis.



Tail-driven sensitivity

The sensitivity to anomalous τ dipole couplings is tail-driven. Relative to the LHeC benchmark, LH μ C combines a larger event sample with a stronger high- $M_{\tau\tau}$ lever arm, making it a promising benchmark for a_τ and d_τ studies.

Current LHC limits on a_τ

CMS exclusive pp photon-induced $\tau^+\tau^-$ production gives $a_\tau \in [-0.0022, 0.0041]$ [2406.03975 [hep-ex]], while the ATLAS high-mass $\tau^+\tau^-$ analysis gives $a_\tau \in [-0.0024, 0.0047]$ (95% CL) [2503.19836 [hep-ex]].

Standard-Model $\gamma\gamma \rightarrow W^+W^-$: rates and energy reach



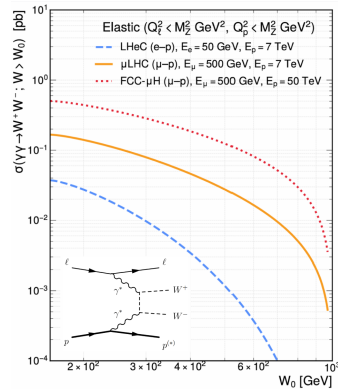
- It is a tree-level electroweak process in the SM.
- The amplitude contains the gauge-related t/u -channel W exchange and the quartic $\gamma\gamma WW$ interaction.
- Production turns on at $W_{\gamma\gamma} \gtrsim 2m_W \simeq 160$ GeV.
- At high energy, vector-boson pair production remains comparatively large:

$\sigma_{\gamma\gamma \rightarrow \nu\nu}(W) \rightarrow$ approximately constant,

unlike the stronger high-energy suppression for fermion and scalar pairs.

- For an illustrative integrated luminosity of 250 fb^{-1} , the elastic SM rates correspond to approximately $N_{WW}^{\text{LHeC}} \simeq 9.3 \times 10^3$ and $N_{WW}^{\text{LH}\mu\text{C}} \simeq 4.2 \times 10^4$ produced W^+W^- pairs, before branching fractions, detector acceptance and reconstruction efficiencies.
- The harder LH μ C photon spectrum therefore provides both larger statistics and a longer high- $W_{\gamma\gamma}$ lever arm for precision studies of the $\gamma\gamma WW$ interaction.

Collider	$\sqrt{s_{\ell p}}$	$\sigma_{\gamma\gamma \rightarrow W^+W^-}^{\text{el}} \text{ [pb]}$
LHeC	1.2 TeV	0.037
LH μ C	3.7 TeV	0.166



Integrated elastic $\gamma\gamma \rightarrow W^+W^-$ production cross section, $\sigma(W_{\gamma\gamma} > W_0)$, as a function of the minimum two-photon invariant mass W_0 for the LHeC, LH μ C and FCC- μ H benchmark configurations.

Anomalous quartic gauge couplings in $\gamma\gamma \rightarrow W^+W^-$

Motivation and EFT description

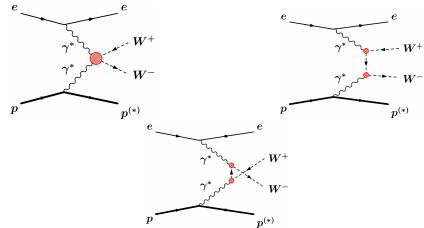
- In the Standard Model, $\gamma\gamma \rightarrow W^+W^-$ is a tree-level electroweak process directly sensitive to the $\gamma\gamma WW$ interaction.
- The clean ℓp environment makes this channel a powerful benchmark for testing the gauge structure of the electroweak sector.
- In the SMEFT framework, heavy new physics is parametrised through higher-dimensional operators:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i}{\Lambda^4} \mathcal{O}_i.$$

- A convenient decomposition is

$$\mathcal{L}_{\text{aQGC}} = \sum_j \frac{f_{S,j}}{\Lambda^4} \mathcal{O}_{S,j} + \sum_j \frac{f_{M,j}}{\Lambda^4} \mathcal{O}_{M,j} + \sum_j \frac{f_{T,j}}{\Lambda^4} \mathcal{O}_{T,j}.$$

- In our benchmark study, the representative deformation f_{M2}/Λ^4 is used to illustrate the sensitivity of $\gamma\gamma \rightarrow W^+W^-$ to anomalous quartic interactions.
- aQGC effects are typically small near threshold, but they grow in the high- M_{WW} and high- p_T tails.



Representative diagrams contributing to $\gamma\gamma \rightarrow W^+W^-$, including the quartic $\gamma\gamma WW$ interaction.

Why this channel is powerful

- Multiple decay channels can be explored: fully leptonic, semileptonic and fully hadronic.
- Both elastic and proton-dissociative production can contribute.
- The observables most sensitive to aQGCs are M_{WW} and p_T^{WW} .
- The LH μ C combines a harder $\gamma\gamma$ spectrum with a clean experimental environment, improving the reach relative to the LHeC.

$\gamma\gamma \rightarrow W^+W^-$: a clean electroweak laboratory



Anomalous quartic gauge couplings in $\gamma\gamma \rightarrow W^+W^-$ 

Heavy final states: $\gamma\gamma \rightarrow t\bar{t}$



What does two-photon physics require from the $\text{LH}\mu\text{C}$ experiment?



- Wide central tracking and lepton acceptance, including robust τ reconstruction.
- Efficient $W/Z/t$ reconstruction for heavy $\gamma\gamma$ final states.
- Exclusivity and acoplanarity observables with sensitivity to low extra activity.
- Missing-momentum performance for neutrinos and compressed BSM signatures.
- Scattered-muon measurement to constrain the lepton-side photon kinematics.
- Forward proton tagging where feasible, together with control of proton dissociation.
- Exploit the defining advantage: the clean, low-pile-up μp environment.

Physics–detector connection

The $\gamma\gamma$ programme links central reconstruction, lepton tagging and near-beam instrumentation directly to the usable photon luminosity and kinematic closure.

Summary and outlook



- LH μ C turns μp collisions into a **high-energy $\gamma\gamma$ laboratory** with a substantially harder photon spectrum than the LHeC benchmark.
- EPA and $S_{\gamma\gamma}$ connect parent μp luminosity transparently to elastic and proton-dissociative two-photon subprocesses.
- $\tau^+\tau^-$ and W^+W^- provide high-statistics precision/EFT channels; sensitivity lives in their hard invariant-mass tails.
- Loop-induced ZZ , exclusive $t\bar{t}$, light-by-light scattering, H , and charged BSM pairs extend the programme to rare quantum processes and heavy thresholds.

Bottom line: high-energy $\gamma\gamma$ physics is one of the distinctive physics opportunities of LH μ C.



The HELP4CERN collaboration



HELP4CERN collaboration focuses on the physics case for high-energy, high-luminosity lepton-hadron colliders at CERN.

- assess the physics potential of future lepton-hadron collider options,
- develop benchmark studies for Higgs, electroweak, top, QCD, and BSM physics,
- provide a common framework for phenomenology, detector studies, and future projections,
- new contributions and physics studies are very welcome,

Contact: help-4-cern@cern.ch

Visit: <https://help4cern.web.cern.ch>

White Paper on High Energy, High Luminosity Lepton-Hadron Colliders at CERN



D. Acosta^a D. Britzger^b A.C. Canbay^c K. Cheung^{d,e,f} B. Dagli^g L. Forthomme^h
 B. Holzerⁱ U. Kaya^j H. Khanpour^h P. Kotko^h M. Kumar^h B. Mellado^h S. Morettiⁱ
 M. Otani^{im} K. Piotrkowski^h D. Schulteⁱ A.M. Staśto^h S. Sultansoy^g I. Tapan^o
 S. Tentori^p M. Topp-Mugglestone^q Z.S. Wang^r Y. Yamazaki^s C. Zhang^t G. Ziarko^h
 F. Zimmermannⁱ

Next steps for HELP4CERN:

- turn the current benchmark studies into a coherent white-paper assessment of high-energy lepton-hadron options at CERN by the end of 2026;
- foster a worldwide study effort, in close coordination with the LHeC Study Group, aimed at preparing a dedicated **HELP4CERN proposal** by 2028.